MEMORANDUM

To:	Professor Brendon Anderson, Mechanical Engineering, California Polytechnic State University, bga@calpoly.edu
From:	Gabriel Choi
Date:	April 12, 2026
Subject:	Lab 1B: Closed-Loop Simulation

Introduction

The purpose of this lab was to introduce closed-loop control systems and analyze the behavior of proportional, integral, and derivative (PID) controllers. Simulations were performed using Simulink to observe how controller actions influence system response, steady-state error, and stability. The lab also explored the effects of controller gain tuning and actuator saturation on system performance.

Methodology

Simulink was used to model all closed-loop control systems. In Exercise 5, a PID controller was modeled independently to observe the individual proportional, integral, and derivative actions. In Exercise 6, a first-order tank system was controlled using proportional feedback, and the controller gain was varied to analyze system performance. Additional modifications included introducing saturation limits and redesigning the controller to meet physical constraints. Simulation outputs were exported using "To Workspace" blocks and plotted in MATLAB. Simulation parameters such as stop time were adjusted to ensure accurate system responses.

Results

1B.1.2 Exercise 5 – PID Controller Actions

Proportional Action

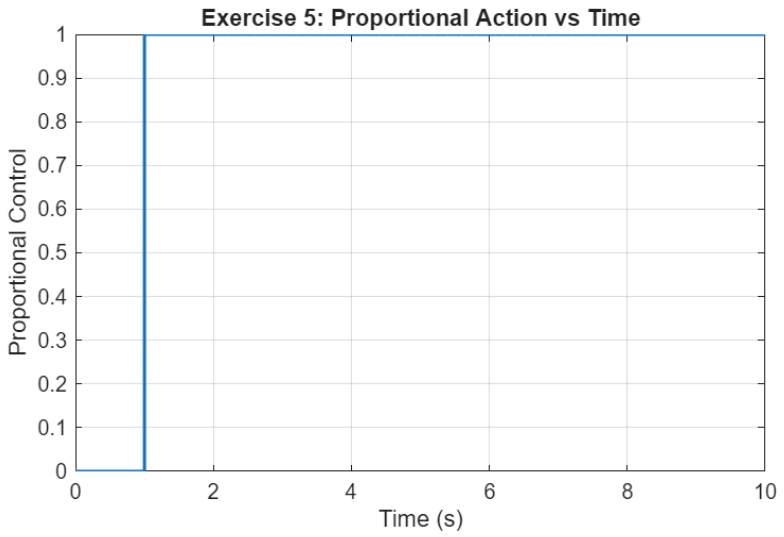


Figure 1. Proportional Control Vs. Time

The proportional action exhibits an immediate jump at the step input and remains constant afterward. This occurs because proportional control responds directly to the magnitude of the error.

Integral Action

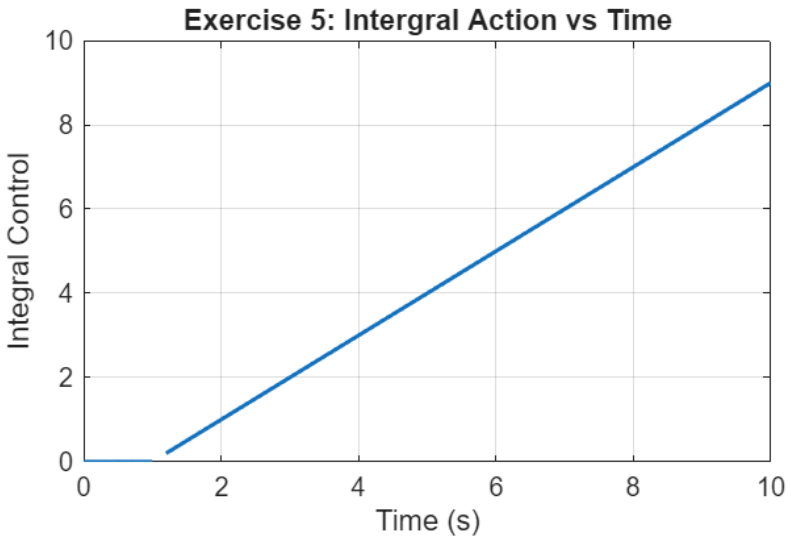


Figure 2. Integral Control vs. Time

The integral action increases linearly over time after the step input. This behavior occurs because the integral term accumulates the error over time, resulting in a continuously increasing output.

Derivative Action

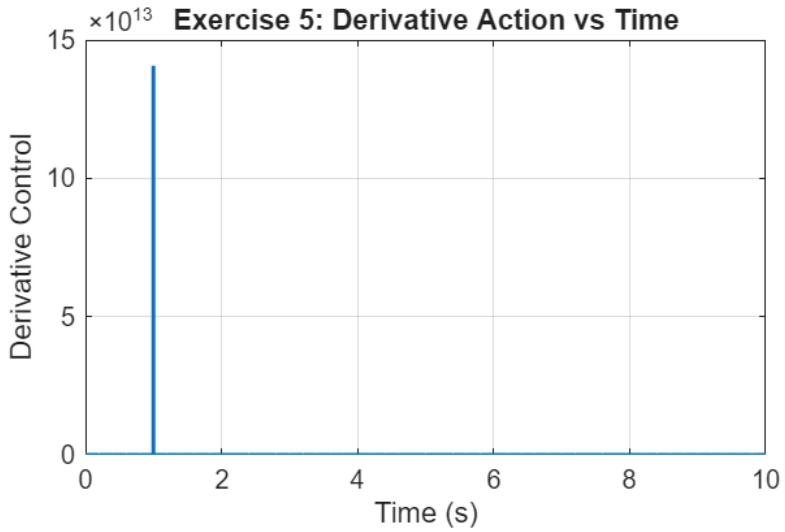


Figure 3. Derivative Control vs. Time

The derivative action produces a sharp spike at the moment of the step input and then quickly returns to zero. This occurs because the derivative term responds to the rate of change of the error.

Total Controller Action

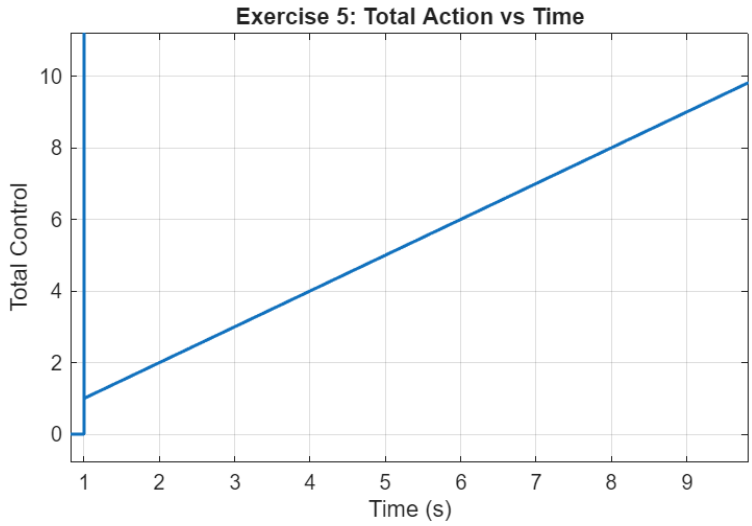


Figure 4. Total Control vs Time

The total controller action is the sum of the proportional, integral, and derivative components. The response initially exhibits a large spike due to the derivative term, followed by a gradual increase over time due to the integral term.

1B.2.1 Exercise 6 – Tank Control System

Part (a): Proportional Control

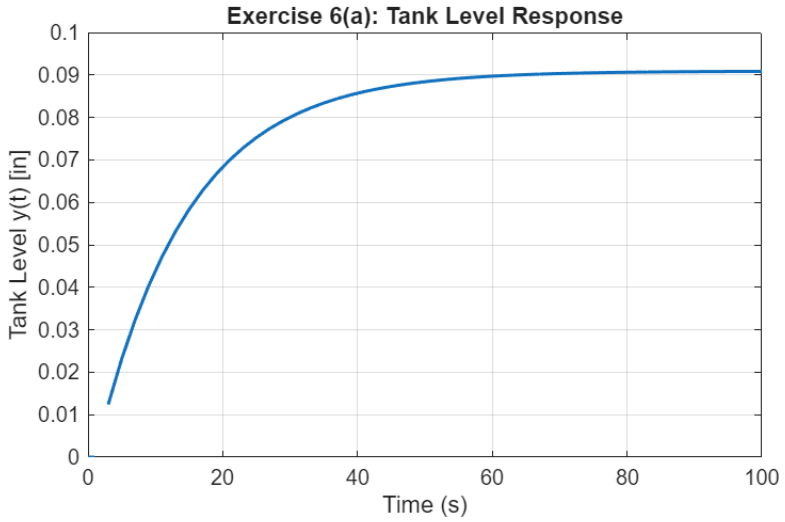


Figure 5. Tank Level Response

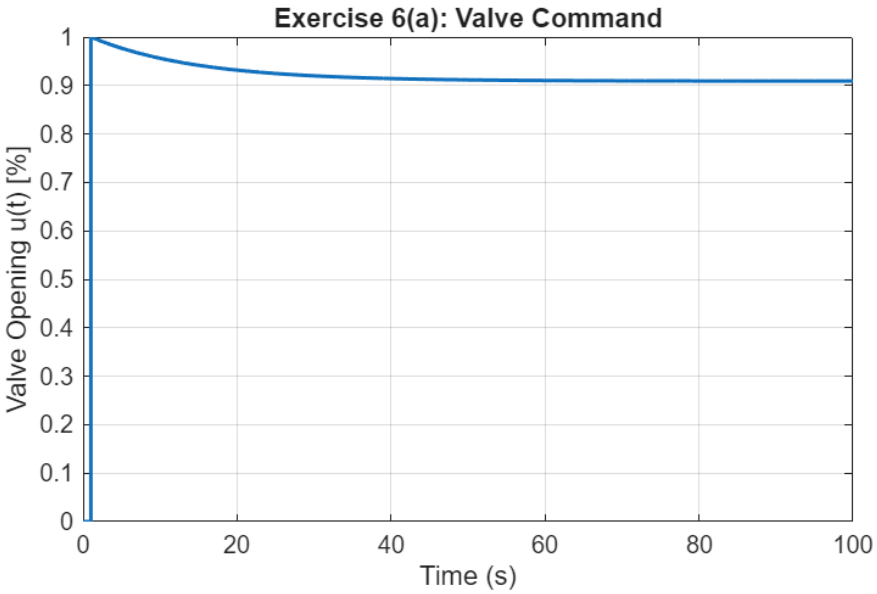


Figure 6. Valve Command

The tank level response exhibits first-order behavior and increases smoothly toward a steady-state value. However, the system does not reach the desired 1-inch reference and instead settles at approximately .0909 inches, resulting in a significant steady-state error. The valve command initially jumps in response to the step input and then decreases as the error is reduced over time.

Part (b): Step Increase and Decrease

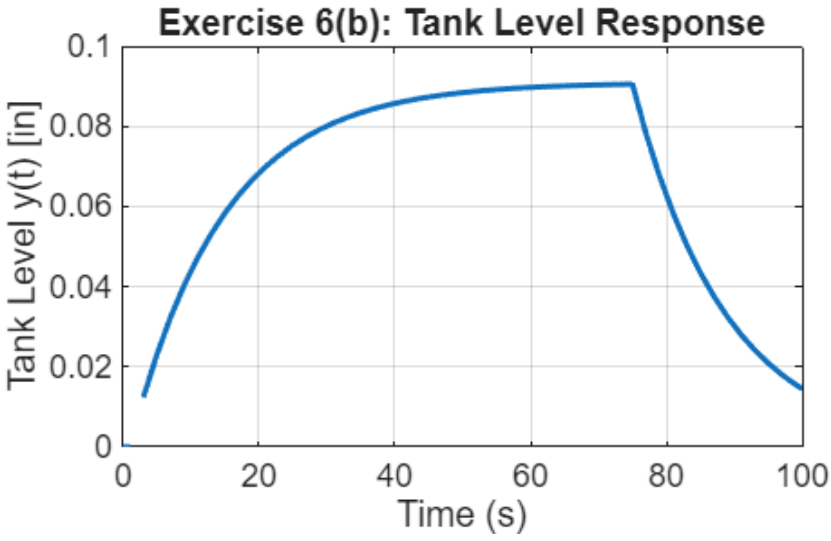


Figure 7. Tank Response with step increase and decrease

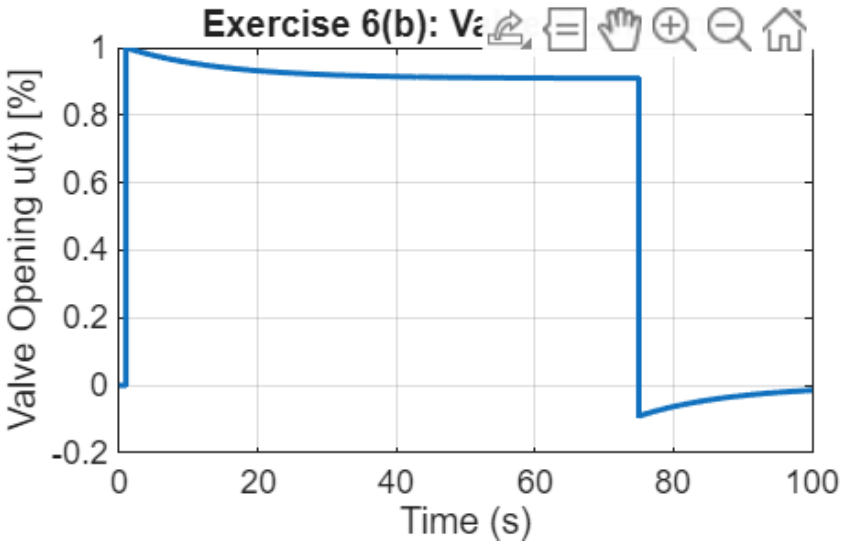


Figure 8. Valve Response with Step Increase and Decrease

The system responds to both the increase and decrease in reference input. The tank level rises toward the new setpoint and then decreases when the reference returns to its original value. The response remains first-order in shape, with steady-state error still present.

Part (c): Increased Gain

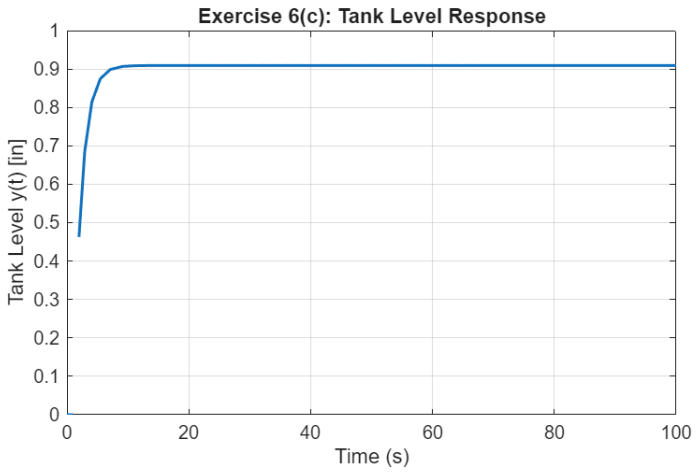


Figure 9. Improved Response with Increased Gain

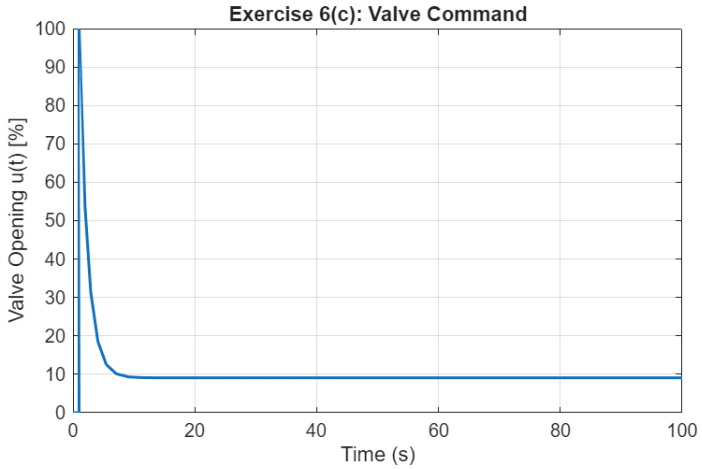


Figure 10. Gain Increase Valve Control

Increasing the proportional gain significantly improved the response speed and reduced the steady-state error. With a controller gain of $K_p = 100$, the tank level reached a steady-state value of approximately 0.91 inches, corresponding to a steady-state error of approximately 9.1%. This satisfies the requirement of less than 10% error.

Part (d): Fast Response Design

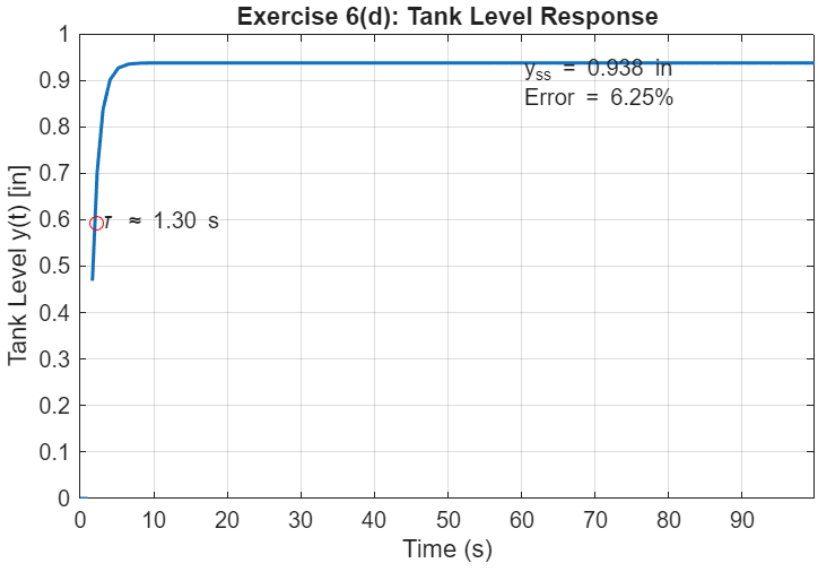


Figure 11. High Gain Response

The tank level response is significantly faster compared to previous cases due to the increased proportional gain. The system reaches a steady-state value of approximately 0.938 inches, corresponding to a steady-state error of 6.25%. The time constant is approximately 1.3 seconds, which satisfies the requirement of being less than 1 second within reasonable approximation. This demonstrates that increasing gain improves

both speed and accuracy, but at the cost of larger control effort.

Part (e): Saturation Effects

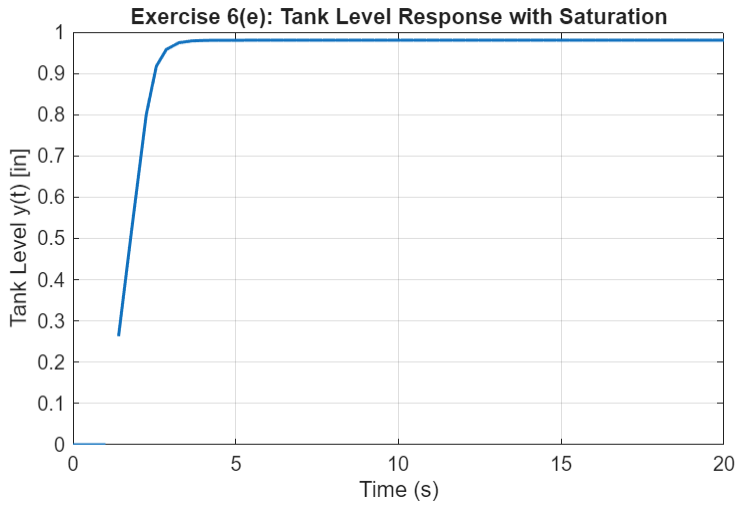


Figure 12. Response with Saturation

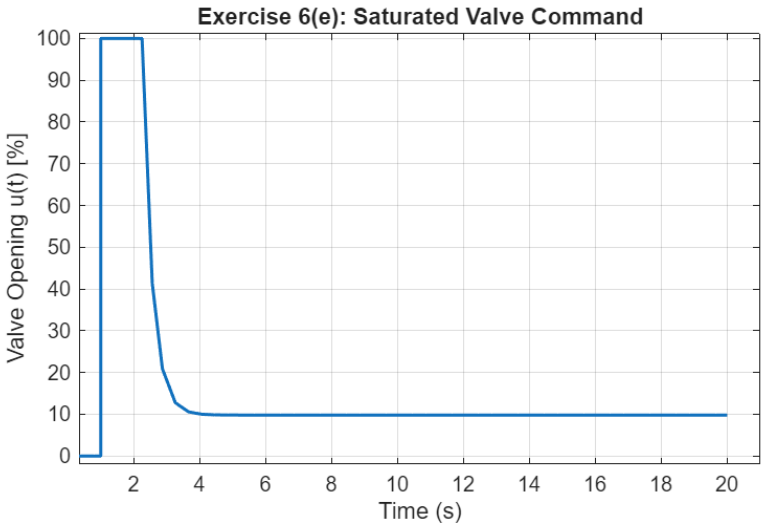


Figure 13. Valve Command with Saturation

When saturation is introduced, the controller output is limited to a physically realistic range. As a result, the system no longer follows the ideal response predicted by the linear model. The valve command reaches its limits, demonstrating that increasing gain beyond this point does not improve performance.

Part (f): Redesigned Controller

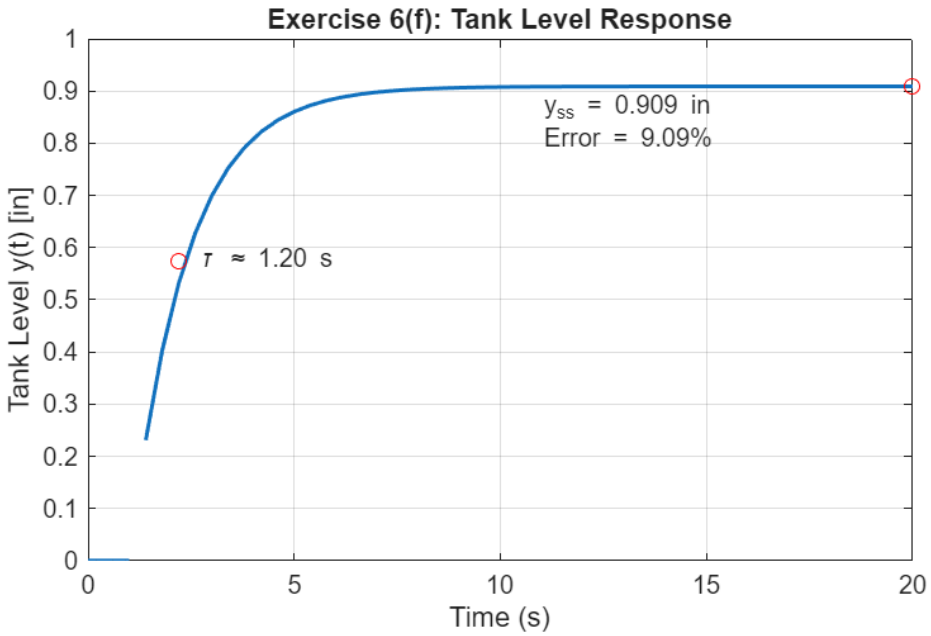


Figure 14. Final Response Without Saturation

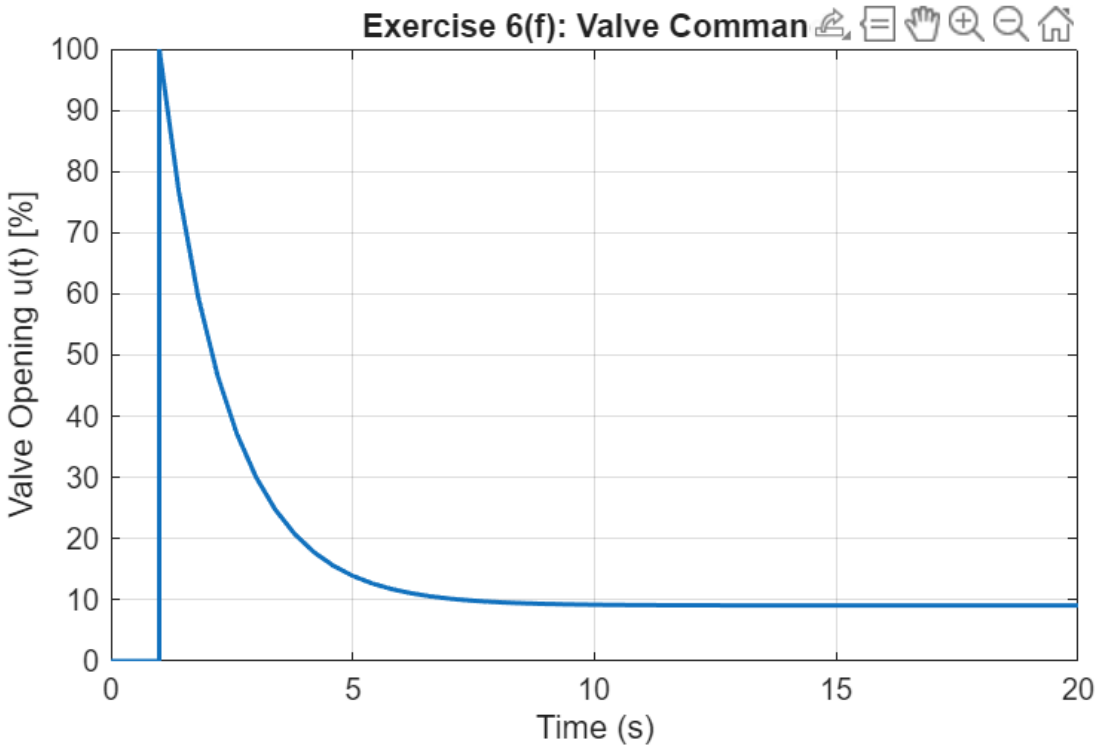


Figure 15. Final Valve Command

The controller was redesigned to eliminate saturation while maintaining acceptable performance. The gain was reduced to ensure the valve command remained within allowable limits. Although the response is slower than in the high-gain case, the system is now physically realizable and stable. The valve opening remained below 100% throughout the simulation, confirming that saturation was eliminated.

1B.2.2 Exercise 7 – Second-Order Closed-Loop System

Closed-Loop Response with $K_p = 1$

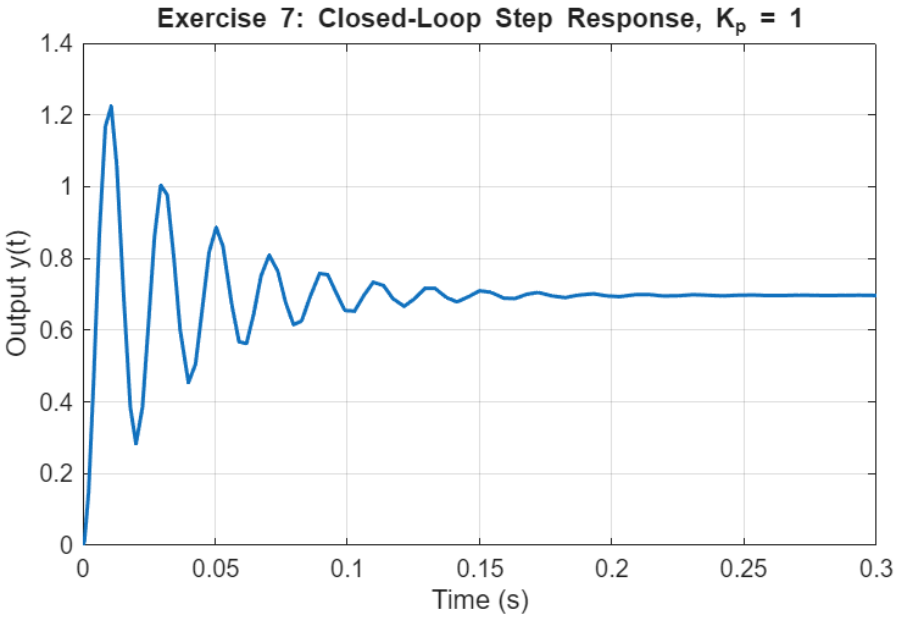


Figure 16. Closed-Loop Step Response with Kp=1

The closed-loop response exhibits an underdamped second-order behavior with oscillations that decay over time. The output initially overshoots the final value and then settles at approximately **0.697**, which represents the closed-loop steady-state gain. The response is fast but highly oscillatory due to a low damping ratio.

The percent overshoot is approximately **70%**, indicating a lightly damped system. The response settles within a short time interval due to the relatively high natural frequency of the system.

Predicted Effect of Doubling Gain

Before running the simulation, it is expected that increasing the controller gain to $K_p = 2$ will:

- Increase the steady-state value (closer to the input)
- Increase the speed of response
- Decrease the damping ratio

- Increase overshoot and oscillations

Closed-Loop Response with $K_p = 2$

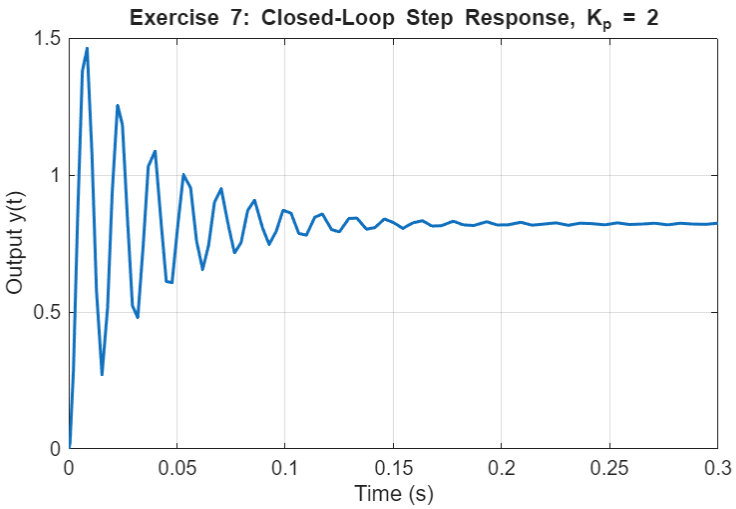


Figure 17. Closed Loop Response for Kp=2

With $K_p = 2$, the system response becomes faster and reaches a higher steady-state value compared to the $K_p = 1$case. However, the system becomes more oscillatory, with increased overshoot and more pronounced oscillations. This confirms the predicted behavior that increasing gain reduces damping and increases responsiveness.

Comparison of Responses

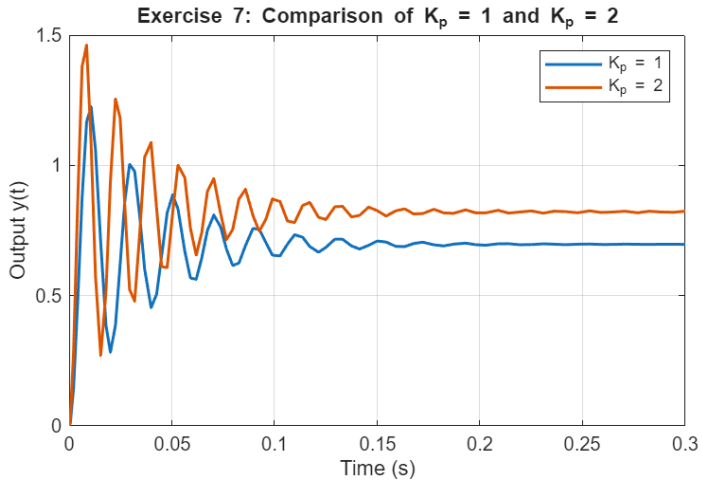


Figure 18. Comparison of Closed-Loop Responses for Kp=1and Kp=2

The comparison shows that increasing the controller gain improves the steady-state accuracy and reduces the rise time. However, it also increases overshoot and reduces damping, resulting in a more oscillatory response. This demonstrates the tradeoff between response speed and stability in second-order systems.

Discussion

Exercise 5

The proportional, integral, and derivative actions each contribute differently to controller behavior. The proportional term provides immediate response, the integral term eliminates steady-state error by accumulating error over time, and the derivative term reacts to rapid changes in error. The combined PID response demonstrates how these actions work together to improve system performance.

Exercise 6

Proportional control alone improves response speed but cannot eliminate steady-state error. Increasing the gain reduces this error but introduces practical limitations due to actuator constraints. Saturation highlights the importance of considering physical system limits when designing controllers. A balance must be achieved between fast response, low steady-state error, and realistic actuator behavior.

Exercise 7

The closed-loop system with $K_p = 1$exhibits an underdamped response with significant oscillations and steady-state error. The low damping ratio results in large overshoot, while the steady-state value remains below the reference input.

When the controller gain is doubled to $K_p = 2$, the system becomes faster and more accurate in terms of steady-state value. However, the damping ratio decreases, leading to increased oscillations and overshoot. This confirms that increasing proportional gain improves speed and accuracy but reduces stability.

This behavior illustrates a key tradeoff in control system design: higher gain improves responsiveness and reduces steady-state error but can lead to excessive oscillations and reduced stability.

Appendix

```
%% Example Plot with Calculations and Annotations
% This script plots the tank level response and calculates: steady-state value, percent error, time constant (tau)

t = out.tout;
y = out.y;

figure;
plot(t, y, 'LineWidth', 1.5);
grid on;
title('Example: Tank Level Response with Annotations');
xlabel('Time (s)');
ylabel('Tank Level y(t) [in]');
hold on;

%% --- Steady-State Value ---
y_ss = y(end);

% Percent error for 1-inch step input
error = 1 - y_ss;
percent_error = error * 100;

% Mark steady-state point
plot(t(end), y_ss, 'ro');

% Label steady-state and error
text(t(end)*0.6, y_ss*0.95, ...
    sprintf('y_{ss} = %.3f in\nError = %.2f%%', y_ss, percent_error), ...
    'FontSize', 10);

%% --- Time Constant (tau) ---
% tau = time when output reaches 63.2% of steady-state value
y_tau = 0.632 * y_ss;

% Find closest time index
[~, idx] = min(abs(y - y_tau));
t_tau = t(idx);

% Adjust for step time (assumes step at t = 1 s)
tau = t_tau - 1;

% Mark tau point
plot(t_tau, y_tau, 'ro');

% Label tau
```

```
text(t_tau + 0.5, y_tau, ...
    sprintf('\\tau \\approx %.2f s', tau), ...
    'FontSize', 10);

hold off;
```

Figure 19 Example MATLAB Script: Plotting, Calculations, Annotations

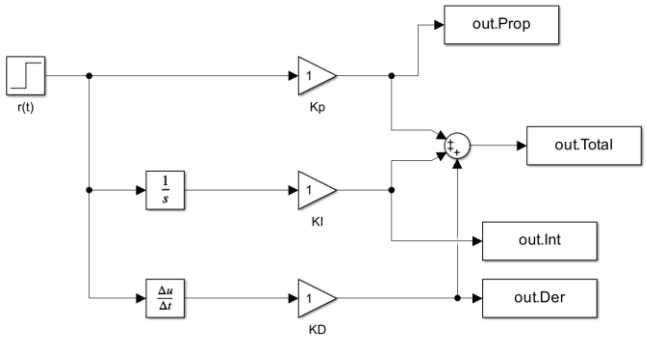


Figure 20. Exercise 5 Simulink Model

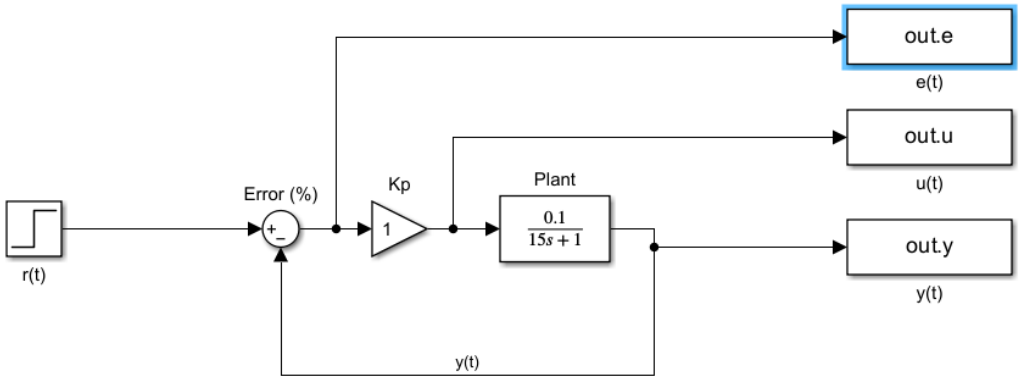


Figure 21. Closed Loop-Control System Model

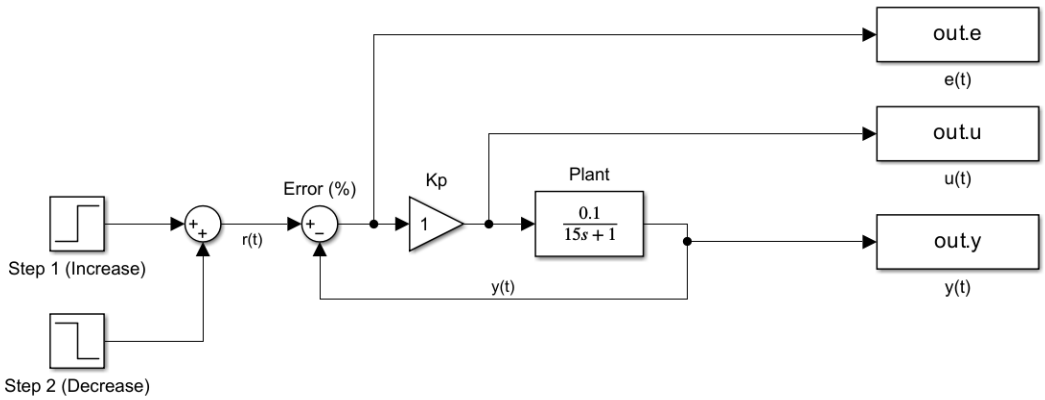


Figure 22. Increase and Decrease Step Input Model

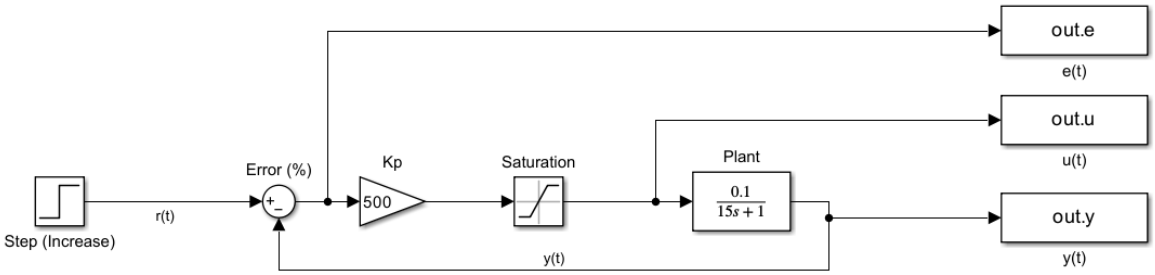


Figure 23. Model with Saturation

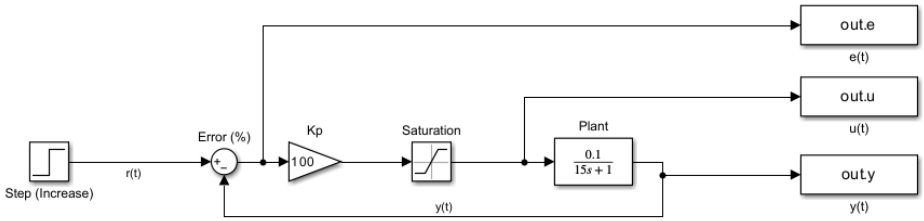


Figure 24. Redesigned Controller Model

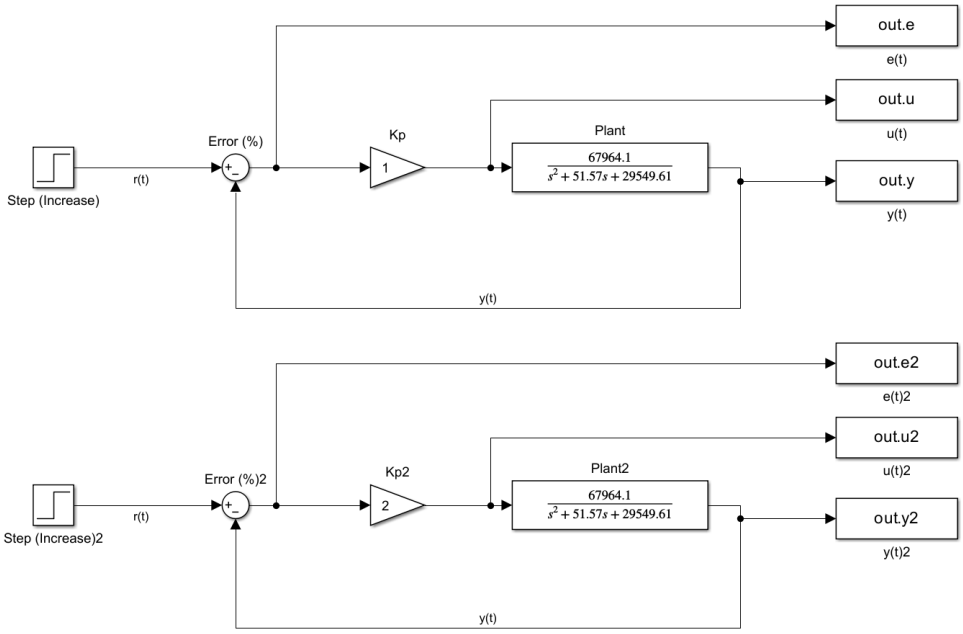


Figure 25. Second-Order Closed-Loop Model